small area of the world. It lasts only for a few minutes. **A partial solar eclipse** (Figure 2.14) happens when the moon partially covers the disc of the sun. **An annular solar eclipse** occurs when the moon passes centrally across the solar disc. During a **total solar eclipse**, the moon's shadow is short enough to cover the whole sun. The outer regions still glow and look bright as a ring. Such a phenomenon is called **Diamond Ring**.

(b) Lunar Eclipse

It occurs on a Full Moon position when the earth is between the sun and the moon. The earth's shadow obscures the moon as viewed from the earth. **A partial lunar eclipse** can be observed when only a part of the moon's surface is obscured by earth's umbra (Figure 2.15). **A penumbral lunar eclipse** happens when the moon travels through the faint penumbral portion of the earth's shadow. **A total lunar eclipse** occurs when the earth

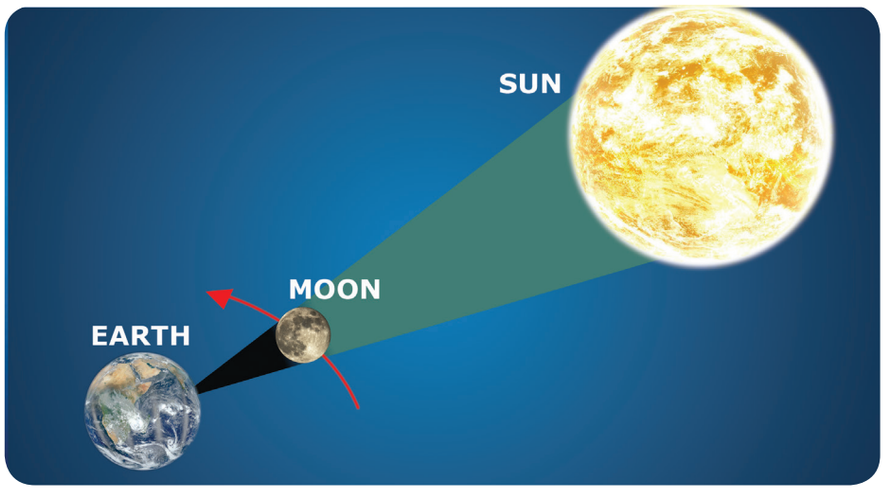


Figure 2.14 Solar eclipse

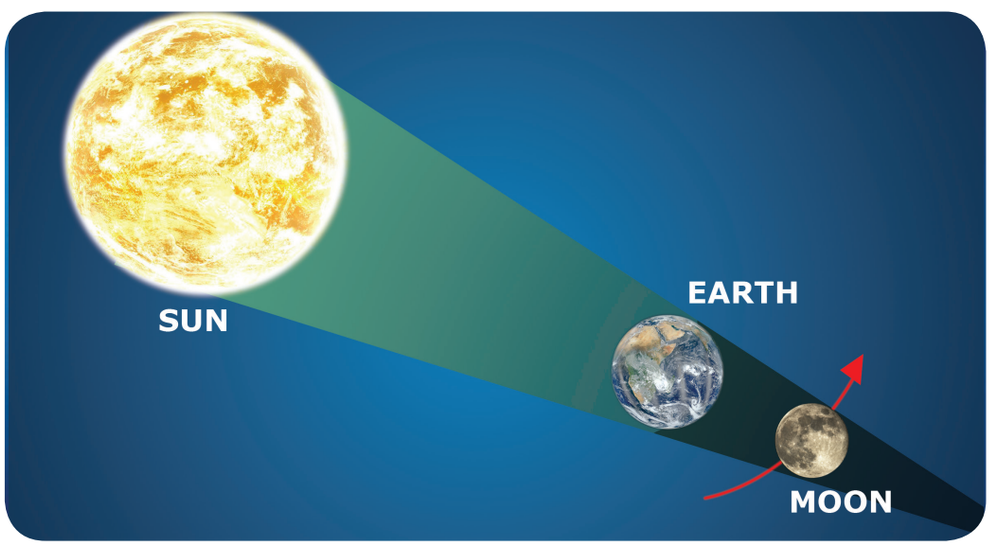


Figure 2.15 Lunar eclipse

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